Claire Najjuuko

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PROFILE

I am a third-year Ph.D. student in the Computational and Data Sciences Program at McKelvey School of Engineering, Washington University in St. Louis. My research focuses on leveraging advanced computational methods, including machine learning predictive models and large language models, to drive innovation in healthcare delivery and improve public health outcomes. I am particularly interested in applications global child health, HIV care, healthcare-associated infections surveillance, and postoperative complications. My research additionally focuses on uncertainty quantification methods for rigorous risk stratification and actionable decision support in real healthcare settings.

EDUCATION

Washington University in St. Louis — St. Louis, MO

Ph.D. in Data and Computational Sciences — September 2023 – Present

- Dean's Select Ph.D. Fellowship (2023–2029)

Makerere University — Kampala, Uganda

Bachelor of Science in Telecommunications Engineering — August 2016 – August 2021

- Government of Uganda Undergraduate Scholarship (2016–2021)

RESEARCH EXPERIENCE

Graduate Student | Washington University in St. Louis September 2023 – Present

Primary research focuses on applications of machine learning and large language models to improve healthcare delivery and public health outcomes.

Project 1: Monitoring HIV Treatment Adherence Using Machine Learning

- **Machine Learning-Based Risk Prediction for ART Adherence (Published):** Developed and evaluated predictive models to identify adolescents at high risk of suboptimal antiretroviral therapy (ART) adherence in low-resource settings. Applied ensemble methods (Random Forest, Gradient Boosting, XGBoost) to socio-behavioral and psychosocial data from 702 adolescents living with HIV in Uganda, achieving AUC of 0.71. Models identified key predictors including depression symptoms, stigma experiences, and caregiver support to support targeted interventions in resource-limited HIV care programs. Published in *AIDS* (2025). Featured in McKelvey News, (Mar 2025).
- **AI-Based Adherence Monitoring Using Clinical LLMs (In Progress):** In this project, we aim to develop large language model-based system for automated adherence monitoring using routinely collected health record data in HIV clinics. Implementing natural language processing and LLM pipelines to extract adherence indicators from clinical notes and using these features to detect ART non-adherence patterns early, enabling timely clinical intervention in resource-limited settings.

Project 2: Machine Learning for Substance Use and HIV Risk Monitoring

- **Predictors of Illicit Substance Use in Young Adults with HIV (Published):** Applied machine learning to longitudinal cohort data from 702 young adults living with HIV in low-resource Ugandan communities to identify key predictors of problematic substance use. Used feature importance analysis and SHAP values to interpret model predictions, identifying behavioral, psychosocial, and structural risk factors including depression, household instability, and peer influence. Findings inform targeted screening and prevention strategies within HIV care programs. Published in *AIDS and Behavior* (2025).
- **UMAT-R AI: Multimodal Prediction of Substance Use Recovery Outcomes (In Progress):** In this study, we are designing an AI pipeline that integrates baseline

surveys, app engagement metrics, and coach-client text messages from the UMAT-R digital intervention to enable early prediction and support more personalized recovery and care. We operationalize the primary outcome as past 30-day substance use recovery and develop text-only and multimodal predictive models using TF-IDF/Word2Vec and clinical BERT variants (e.g., ClinicalBERT, BioClinicalBERT), combined with survey features. Ongoing works include extracting participant-level themes via transformer-based topic modeling (BERTopic), as well as self-supervised and supervised adaptation of transformer encoders on UMAT-R messages.

Project 3: Hybrid AI Systems for Infection Surveillance and Clinical Workflows

- **Hybrid LLM + Rules-Based CAUTI Surveillance (Published):** Developed and validated hybrid surveillance system combining rule-based NHSN logic with LLM-driven symptom extraction from clinical notes for catheter-associated urinary tract infection (CAUTI) surveillance. I implemented the Clinical Entity Augmented Retrieval (CLEAR) framework using BIOMed_NER for entity recognition, BioClinicalBERT embeddings for semantic similarity, and GPT-4o for symptom extraction from flagged potential CAUTI cases at BJH (2021–2024). Our hbrid system achieved 90.0% sensitivity and 93.5% specificity, outperforming rules-based approaches (80.4% sensitivity) and standalone LLM methods (88.7% sensitivity), while reducing infection preventionist chart review workload by approximately 66%. I managed the complete data pipeline including data processing, model development, evaluation, and error analysis. I am Co-first author, in the article published in *Clinical Infectious Diseases* (2026).Also, featured in Division of Infectious Diseases News, (May 2026).

Project 4: Quantifying Uncertainty in Clinical Prediction Models

- **Uncertainty-Aware AI for OR-to-ICU Handoff Decision Support (In Progress):** Developing and evaluating an uncertainty estimation framework for AI-predicted postoperative complications (acute kidney injury, delirium, deep vein thrombosis, pulmonary embolism, and pneumonia) to support OR-to-ICU handoff decision-making. The framework uses a two-stage conformal prediction approach: in the first stage, a conformal predictor generates preliminary prediction sets with a coverage guarantee; in the second stage, confident singleton rule-outs (where the model predicts {Low Risk} alone) are re-evaluated against similar patients in the calibration set. If the rule-out is not well-supported, the prediction is expanded to {Low Risk, High Risk}, signaling uncertainty to the clinician rather than issuing a potentially dangerous confident rule-out. Preliminary results show that this two-stage approach reduces false negative rates across all five complication models. A within-subjects user study will evaluate which uncertainty communication format best supports handoff decisions.
- **Conformal Prediction for Emergency Medical Services (In Progress):** Building and evaluating conformal prediction-based models for emergency medical services (EMS) that predict intervention protocols, medication types, and quantities from multimodal data (free-text symptoms + vital signs). Developing calibrated, class-fair prediction sets and selective-classification strategies to provide trustworthy decision support for high-stakes clinical interventions.
- **Uncertainty Quantification Methods for Clinical Language Models (In Progress):** In this study we are conducting a benchmark study characterizing uncertainty quantification methods for clinical language models, evaluating their utility and reliability in real-world clinical prediction settings. We compare approaches including Monte Carlo dropout, ensemble methods, and calibration techniques across multiple clinical tasks.

Project 5: Maternal and Child Health Service Utilization (Published)

- Analyzed patterns of maternal and child health services utilization and associated socioeconomic disparities across sub-Saharan Africa using machine learning approaches on Demographic and Health Survey data from 352,363 mother-child pairs across 35 countries. Identified critical service utilization patterns linked to child mortality outcomes. Published in *Nature Communications* (2025). Featured in McKelvey News, August 2025.

Undergraduate Research Assistant | Makerere University **July 2020 – June 2021**

- **Rural Broadband Connectivity Model:** Piloted rural broadband connectivity model funded by Research and Innovations Fund Uganda (PI: Eng. Dr. Dorothy Okello). Conducted technical feasibility studies for 5G and TV white space technologies in rural Ugandan communities. Co-authored two IEEE conference papers on rural connectivity solutions.

Undergraduate Student | Makerere University **August 2019 – December 2020**

- **Final Year Project:** Developed machine learning models to optimize sleep mode in dense cellular networks, implementing energy-efficient algorithms to reduce power consumption while maintaining optimal network performance.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Nordman J*, **Najjuuko C***, Jeschke N, Snyders RE, Vazquez Guillamet MC, Shapiro J, Leone C, Dethloff M, Babcock H, Reich P, Whalen K, Zhang L, Luong L, Lu C, Atkinson A, Marschall J, Sung A. The Best of Both Worlds: How Combining a Large Language Model and a Rule-Based Algorithm Makes CAUTI Surveillance More Efficient. *Clinical Infectious Diseases*. 2026; ciag301. *Co-first authors. https://doi.org/10.1093/cid/ciag301
2. Ssentumbwe V, Matovu F, Namatovu P, **Najjuuko C**, Kizito S, Nabayinda J, Nartey P, Namuwonge F, Nabunya P, Lu C, Mutumba M, Ssewamala FM. Using Machine Learning to Predict Depression among Adolescents Living with HIV in Uganda. *Global Social Welfare*. 2026 Mar 21. https://doi.org/10.1007/s40609-026-00456-3
3. **Najjuuko C**, Brathwaite R, Mutumba M, Childress S, Nannono S, Namatovu P, Lu C, Ssewamala FM. Identifying Predictors of Problematic Substance Use Among Youth Living with HIV in Uganda: A Machine Learning Approach. *AIDS and Behavior*. 2025 Sep 18:1-2. https://doi.org/10.1007/s10461-025-04840-6
4. **Najjuuko C**, Xu Z, Kizito S, Lu C, Ssewamala FM. Patterns of maternal and child health services utilization and associated socioeconomic disparities in sub-Saharan Africa. *Nature Communications*. 2025; 16:7840. https://doi.org/10.1038/s41467-025-61350-8
5. **Najjuuko C**, Brathwaite R, Xu Z, Kizito S, Lu C, Ssewamala FM. Using machine learning to predict poor adherence to antiretroviral therapy among adolescents living with HIV in low resource settings. *AIDS*. 2022 Apr 20:10-97. https://doi.org/10.1097/QAD.0000000000004163
6. Tutlam NT, Nabunya P, Kizito S, Migadde H, Ssentumbwe V, Namuwonge F, **Najjuuko C**, Mugisha J, Sensoy Bahar O, Mwebembezi A, Ssewamala FM. The Impact of Group-Based Interventions on Emotional and Behavioral Difficulties among Adolescents Living with HIV: The Suubi4Stigma Cluster Randomized Controlled Trial. *Journal of Child and Family Studies*. 2025 Feb 12:1-9. https://doi.org/10.1007/s10826-025-03015-0
7. Nabunya P, Ssewamala FM, Kizito S, Mugisha J, Brathwaite R, Neilands TB, Migadde H, Namuwonge F, Ssentumbwe V, **Najjuuko C**, Bahar OS. Preliminary Impact of Group-Based Interventions on Stigma, Mental Health, and Treatment Adherence Among Adolescents Living with Human Immunodeficiency Virus in Uganda. *The Journal of Pediatrics*. 2024 Jun 1;269:113983. https://doi.org/10.1016/j.jpeds.2024.113983
8. Nabunya P, Migadde H, Namuwonge F, Mugisha J, Kirabo W, Ssentumbwe V, **Najjuuko C**, Raymond A, Bahar OS, Mwebembezi A, McKay MM. Feasibility and acceptability of Group-based stigma reduction interventions for adolescents living with HIV and their caregivers: the Suubi4Stigma Randomized Clinical Trial (2020-2022). *AIDS and Behavior*. 2024 Feb 3:1-2. https://doi.org/10.1007/s10461-024-04284-4

9. Kizito S, Namuwonge F, Nabayinda J, Nalwanga D, **Najjuuko C**, Nabunya P, Atwebembere R, Namuyaba OI, Mukasa M, Ssewamala FM. A Cluster-Randomized Controlled Trial of an Economic Strengthening Intervention to Enhance Antiretroviral Therapy Adherence among Adolescents Living with HIV. *AIDS and Behavior.* 2024 May;28(5):1570-80. https://doi.org/10.1007/s10461-024-04268-4
10. Nabunya P, Cavazos-Rehg P, Mugisha J, Kasson E, Namuyaba OI, **Najjuuko C**, Nsubuga E, Filiatreau LM, Mwebembezi A, Ssewamala FM. An mHealth Intervention to Address Depression and Improve Antiretroviral Therapy Adherence Among Youths Living With HIV in Uganda: Protocol for a Pilot Randomized Controlled Trial. *JMIR Research Protocols.* 2024 Mar 8;13(1):e54635. https://doi.org/10.2196/54635
11. White PJ, Okello D, Casey BP, **Najjuuko C**, Lukanga R. Co-designing with engineers for community engagement in rural Uganda. *Design Science.* 2023 Jan;9:e12. https://doi.org/10.1017/dsj.2023.10

Conference Papers

12. Wang S, **Najjuuko C**, Xu Z, Alba C, Lu C. CURA: Clinical Uncertainty Risk Alignment for Language Model-Based Risk Prediction. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2026).* 2026. https://arxiv.org/abs/2604.14651
13. **Najjuuko C**, Ayebare GK, Lukanga R, Mugume E, Okello D. A survey of 5G for rural broadband connectivity. *In 2021 IST-Africa Conference (IST-Africa)* 2021 May 10 (pp. 1-10). IEEE. https://ieeexplore.ieee.org/abstract/document/9576832
14. Lukanga R, Ayebare GK, **Najjuuko C**, Mugume E, Okello D. Leveraging TV white spaces for rural broadband connectivity. *In 2021 IST-Africa Conference (IST-Africa)* 2021 May 10 (pp. 1-8). IEEE. https://ieeexplore.ieee.org/document/9576962

Manuscripts Under Review

15. **Najjuuko C**, Kabarambi A, Ssentumbwe V, Nartey P, Nabunya P, Sensoy Bahar O, McKay MM, Ssewamala FM. The Long-term Impact of a Family Intervention on Symptoms of Attention-Deficit/Hyperactivity Disorder (ADHD) Among Primary School-Going Children in Uganda, 2016-2022. *Under preparation.*

PRESENTATIONS

1. Nordman J*, **Najjuuko C***, Jeschke N, Snyders RE, Vazquez Guillamet MC, Shapiro J, Leone C, Dethloff M, Babcock H, Reich P, Whalen K, Zhang L, Luong L, Lu C, Atkinson A, Marschall J, Sung A. The Best of Both Worlds: How Combining a Large Language Model and a Rule-Based Algorithm Makes CAUTI Surveillance More Efficient. *Poster presentation at the 2026 I^2DB Annual Symposium Innovation*, Washington University in St. Louis, St. Louis, MO, April 2026. Honorable Mention: Research Excellence.
2. **Najjuuko C**, Brathwaite R, Xu Z, Kizito S, Lu C, Ssewamala FM. Using machine learning to predict poor adherence to antiretroviral therapy among adolescents living with HIV in low resource settings. *Project presentation at Mini-Conference on AI and Poverty: Advancing Health, Education, and Equity*, NYU Silver School of Social Work, New York, NY, November 2025.
3. **Najjuuko C**, Brathwaite R, Mutumba M, Childress S, Nannono S, Namatovu P, Lu C, Ssewamala FM. Identification of illicit substance use predictors in young adults living with HIV in low resource communities in Uganda: A machine learning approach. *Poster presentation at Society for Prevention Research (SPR) 33rd Annual Meeting*, Seattle, WA, May 2025. https://spr.confex.com/spr/spr2025/meetingapp.cgi/Paper/34697
4. **Najjuuko C**, Brathwaite R, Xu Z, Kizito S, Lu C, Ssewamala FM. Using machine learning to predict poor adherence to antiretroviral therapy among adolescents living with HIV in low resource settings. *Poster presentation at Global Health at WashU: Focus on Impact*, Washington University in St. Louis, April 2025.

WORK EXPERIENCE

Data Manager | International Center for Child Health and Development Masaka, Uganda
Report to: Principal Investigator September 2021 – July 2023

Study Coordination:

- **mHealth Study:** Coordinated mobile health intervention protocol development to address depressive symptoms among youth living with HIV in Uganda (Funded by McDonnell Scholars Academy; PIs: Proscovia Nabunya, Patricia Cavazos-Rehg).

Data Management (managed data for 6 concurrent longitudinal studies):

- **M-Suubi (R01MH126892):** Multi-level integrated intervention to reduce HIV stigma impact on treatment outcomes among adolescents living with HIV
- **Bridges R2 (R01HD070727-02):** Longitudinal impacts of economic intervention on HIV risk prevention and care continuum outcomes among orphaned youth transitioning to young adulthood
- **Suubi4Stronger Families (R01MH128905):** Addressing child behavioral health by strengthening financial stability and parenting among families in Uganda
- **Suubi4Stigma (R21MH121141):** Addressing HIV-associated stigma among adolescents
- **Suubi+Adherence R2 (R01HD074949):** Longitudinal HIV treatment adherence among youth living with HIV transitioning into young adulthood
- **Kyaterekera (R01MH116768):** 5-year longitudinal study addressing sexual risk-taking behaviors among vulnerable women in Uganda

TECHNICAL SKILLS

Programming and Scripting Languages: Python, R, SQL, LaTeX
Machine Learning & Deep Learning: scikit-learn, TensorFlow, PyTorch
Natural Language Processing & LLMs: HuggingFace Transformers, spaCy, OpenAI API, LangChain, RAG pipelines
Development Tools & Environments: Jupyter, Git/GitHub, Docker, VSCode
Statistical Software: STATA, SPSS

CONFERENCES, WORKSHOPS, AND SEMINARS

- Mini-Conference on AI and Poverty: Advancing Health, Education, and Equity, NYU Silver School of Social Work, November 2025
- Society for Prevention Research (SPR) 33rd Annual Meeting, Seattle, WA, May 2025
- Global Health at WashU: Focus on Impact, Washington University in St. Louis, 2025
- AI for Health Symposium, Washington University in St. Louis, 2024
- AI for Health Symposium, Washington University in St. Louis, 2023
- Forum on Child and Adolescent Global Health Research and Capacity Building, 2023
- Forum on Child and Adolescent Global Health Research and Capacity Building, 2022
- The 6th National Conference on Communications, 2020
- Uganda Institution of Professional Engineers Students Workshop, 2020
- Global Grand Challenges Summit, Kampala Satellite Event, 2019
- Wireless Communication Workshop organized by IEEE Communication Society, 2018

ADDITIONAL TRAINING

- Researcher Resilience Training Program, Fellow 2024
- Researcher Resilience Training Program, Fellow 2023

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